School of Future Tech

Case Study Report

on

FixItNow – Appliance Repair Booking Website

by

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1. Introduction to the Case Study

With the rapid growth of digital services, customers increasingly rely on online platforms to book services conveniently. Traditional appliance repair services often require phone calls, manual scheduling, and unclear pricing, which leads to inconvenience and delays.

This case study focuses on the **design and development of a web-based Appliance Repair Booking System called "FixItNow."** The system allows users to view services, check pricing, view technicians, and book repair services directly through a website.

The project is developed using **HTML, CSS, and JavaScript**, which makes the website interactive, responsive, and user-friendly.

The system demonstrates how front-end web technologies can be used to build a practical real-world service platform.

2. Problem Statement / Case Background (Abstract)

Background

Many local repair services still rely on offline methods of booking appointments. Customers often face problems such as:

- Lack of transparency in service pricing
- Difficulty scheduling repair appointments
- No easy way to check technician availability
- Manual and time-consuming booking processes

These limitations create inefficiency for both customers and service providers.

Abstract

This case study presents the development of **FixItNow**, a web-based appliance repair booking platform that allows users to easily schedule repair services.

The website includes features such as:

- Viewing appliance repair services
- Transparent pricing information
- Technician listings with experience
- Online repair booking form
- Cost estimation based on repair type

The system uses **HTML for structure, CSS for design and layout, and JavaScript for interactive functionality and form validation.**

The project demonstrates how modern web technologies can be used to create a **simple yet effective service booking platform.**

3. Case Study Design

The FixItNow system is designed with multiple web pages and modules.

1. Homepage

The homepage introduces the platform and displays available appliance repair services such as:

- Air Conditioner Repair
- Refrigerator Repair
- Washing Machine Repair
- Microwave Repair

It also highlights key features like:

- Certified technicians
- Transparent pricing
- Same-day service

2. Services Page

This page shows detailed repair services for different appliances and common problems.

Examples of issues handled:

- AC not cooling
- Refrigerator not cooling
- Washing machine not draining
- Microwave not heating

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3. Pricing Page

The pricing page explains how repair costs are calculated.

Example:

Diagnosis Fee + Service Charge + Emergency Fee = Total Cost

This ensures transparency in pricing for customers.

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4. Technicians Page

This page displays available technicians with details such as:

- Technician name
- Years of experience
- Certifications
- Specializations

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5. Booking Page

The booking page allows customers to schedule a repair service.

Users must fill details such as:

- Name
- Phone number
- Address
- Appliance type
- Issue description
- Preferred date and time

The system validates the form before submission.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Technologies Used

1. HTML (HyperText Markup Language)

HTML is used to structure the website pages including:

- Homepage
- Services page
- Pricing page
- Technicians page
- Booking page

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2. CSS (Cascading Style Sheets)

CSS is used for styling and layout of the website.

Key features include:

- Responsive layout

- Navigation bar
- Service cards
- Buttons and forms
- Modern UI design

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3. JavaScript

JavaScript adds interactivity and dynamic functionality.

Main features implemented using JavaScript include:

- Sticky navigation bar
- Mobile menu toggle
- Booking form validation
- Dynamic cost calculation
- Error handling for user inputs
- Confirmation message after booking

Development Tools

The project was developed using:

- **Visual Studio Code**
- **Web Browser (Chrome/Edge)**
- **HTML, CSS, JavaScript**
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    - Matplotlib and Seaborn for visualization of metrics, class distributions, and confusion matrix.
- Environment:
    - Jupyter Notebook, VS Code, or any Python IDE.
- Data storage:

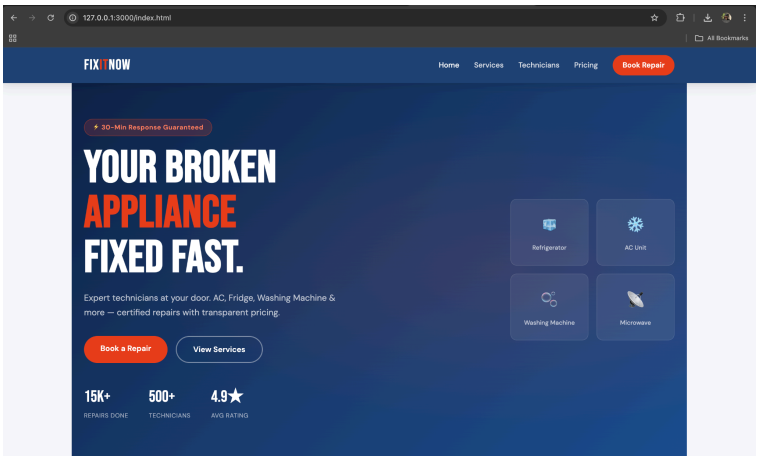
- Dataset stored as CSV or text files.

5. Problem Statement / Case Study Implementation Details and Snapshots.

The website project is divided into several files.

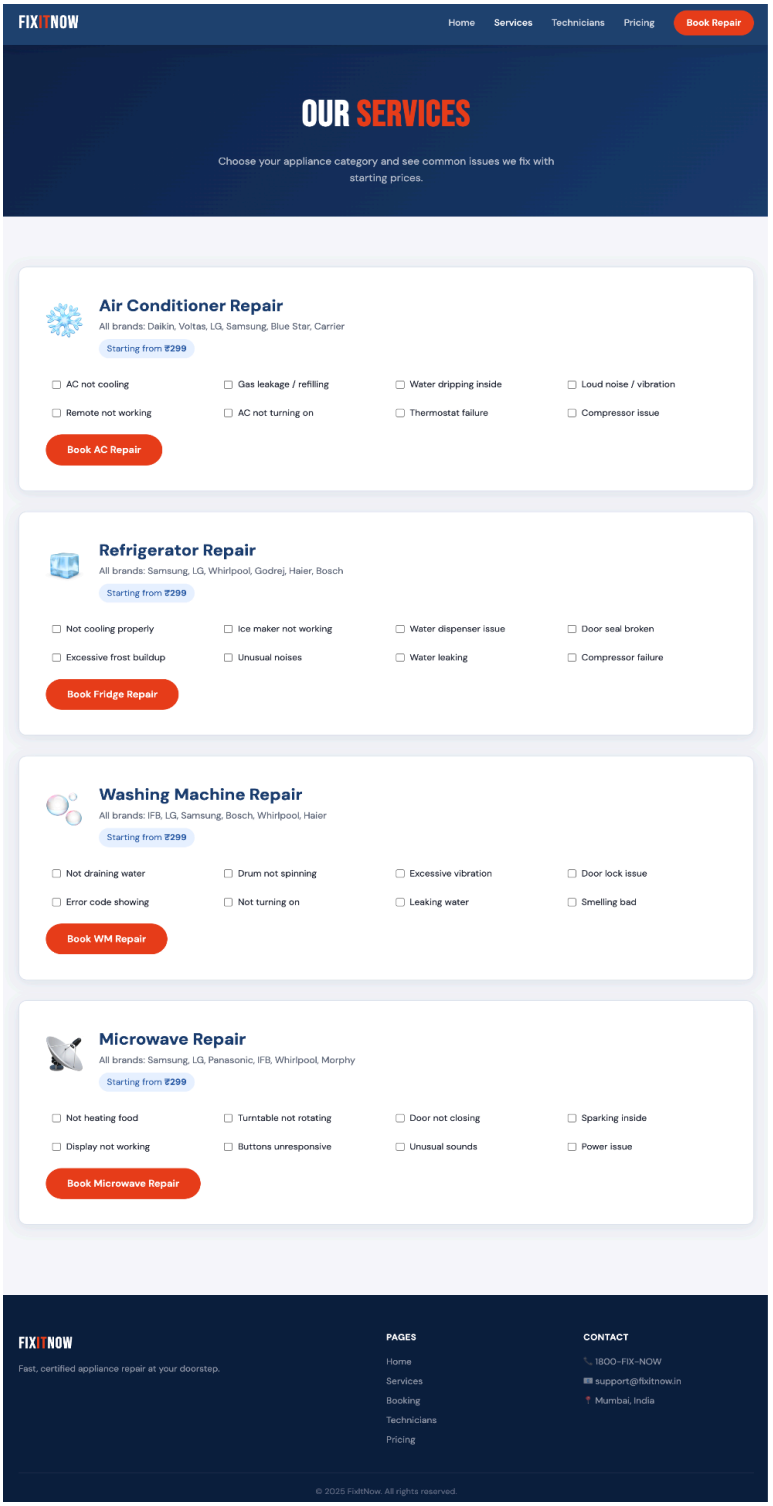
Main Files

1. **index.html**
   Contains the homepage and service overview.



2. **services.html**

Displays detailed repair services.



3. **pricing.html**

Shows service charges and cost

TRANSPARENT PRICING

No hidden fees. Know exactly what you pay before the technician arrives.

🔍 Diagnosis Fee

A flat diagnosis fee is charged for every repair visit, regardless of the appliance or issue.

Service Type	Diagnosis Fee	Includes
Standard Visit	₹299	Inspection + fault report
Emergency Visit (24/7)	₹329	Priority dispatch + inspection
Re-visit (within 30 days)	FREE	Warranty follow-up visit

⚙️ Service Charge Categories

Repair costs are calculated based on issue severity. Charges are applied on top of the diagnosis fee.

Issue Category	Additional Charge	Examples
Minor Issue	₹500	Cleaning, minor parts, sensor reset
Major Issue	₹1,200	Compressor, PCB, motor replacement
Parts (if required)	At Actuals	OEM/compatible parts at MRP

📦 Appliance-Wise Starting Price

Appliance	Minor Repair	Major Repair	Gas Refill / Special
❄️ Air Conditioner	₹799	₹1,499	₹1,200
🧊 Refrigerator	₹799	₹1,499	₹999
🫧 Washing Machine	₹799	₹1,499	N/A
📡 Microwave	₹799	₹1,499	N/A
🚿 Geyser	₹799	₹1,299	N/A

🚨 Emergency Surcharge Details

FIXITNOW

Home Services Technicians Pricing Book Repair

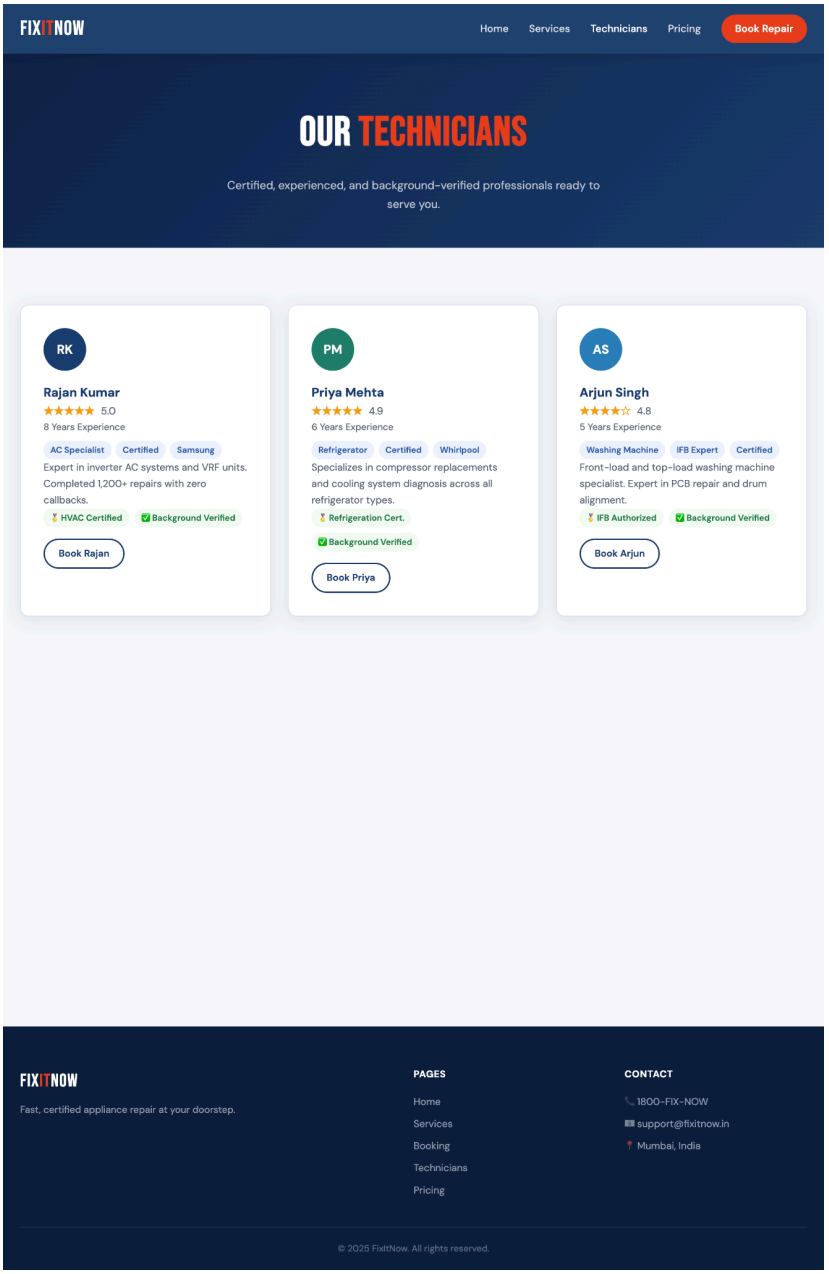
Emergency repairs requested outside standard hours (after 8 PM or before 8 AM), or requiring dispatch within 60 minutes, attract a 10% surcharge on the total repair bill (diagnosis + service charges).

- ✅ Available 24 hours, 7 days a week
- ✅ 60-minute guaranteed dispatch
- ✅ Priority senior technician assigned
- ⚠️ Surcharge waived for Premium members

calculations.

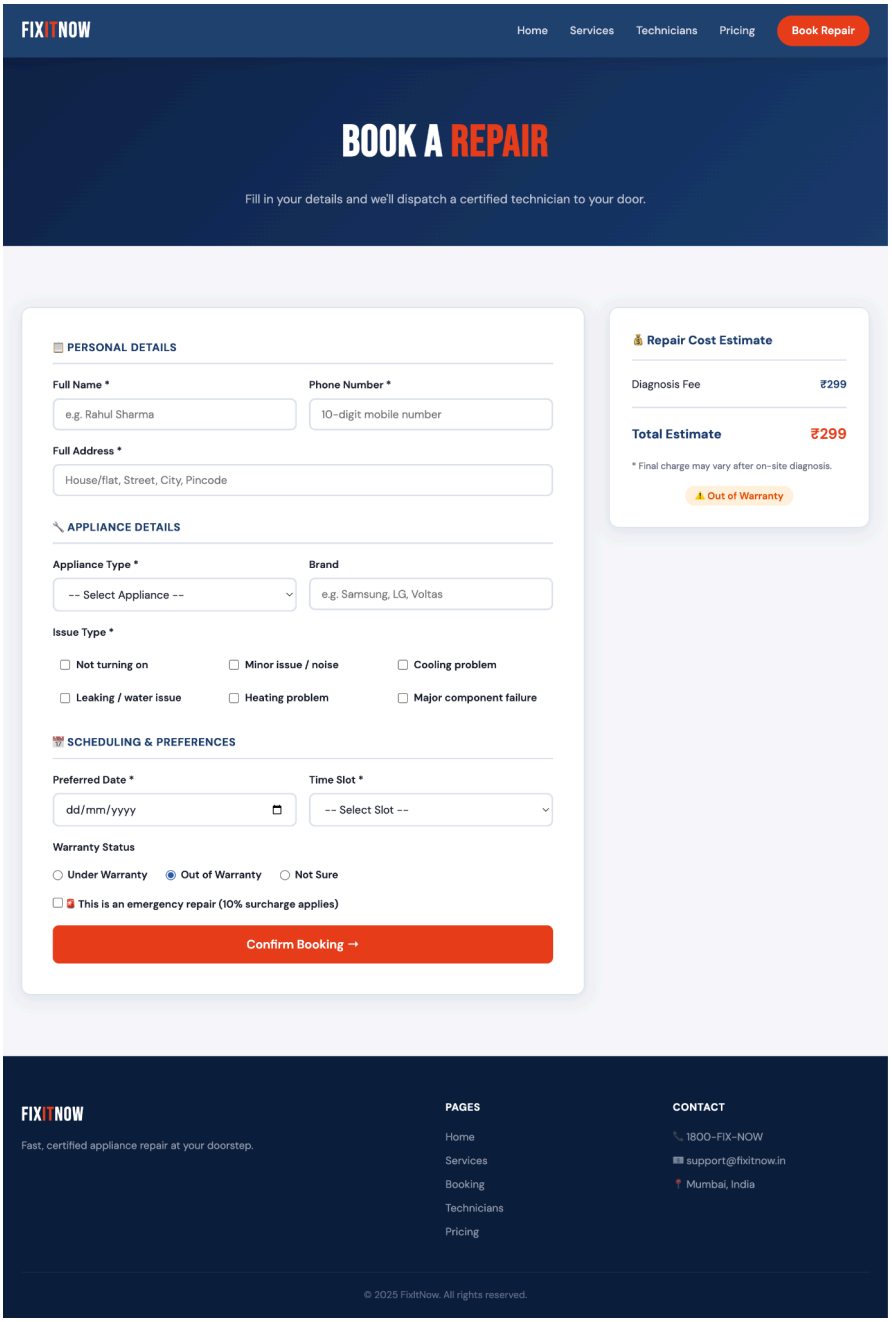
4. **technicians.html**

Lists available technicians and their details.



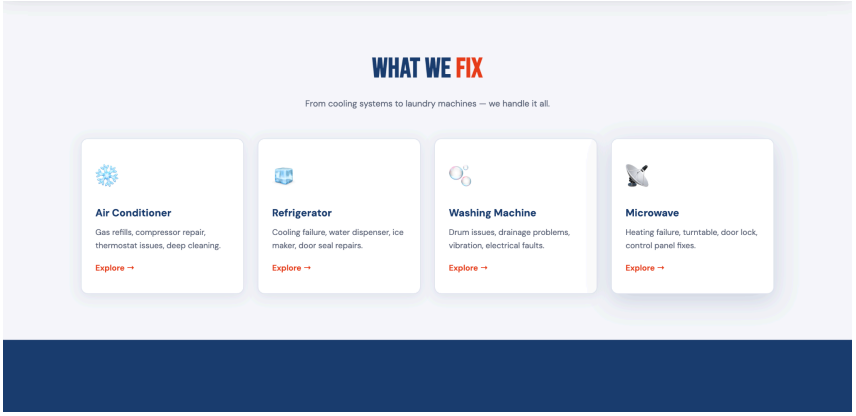
5. **booking.html**

Contains the repair booking form.



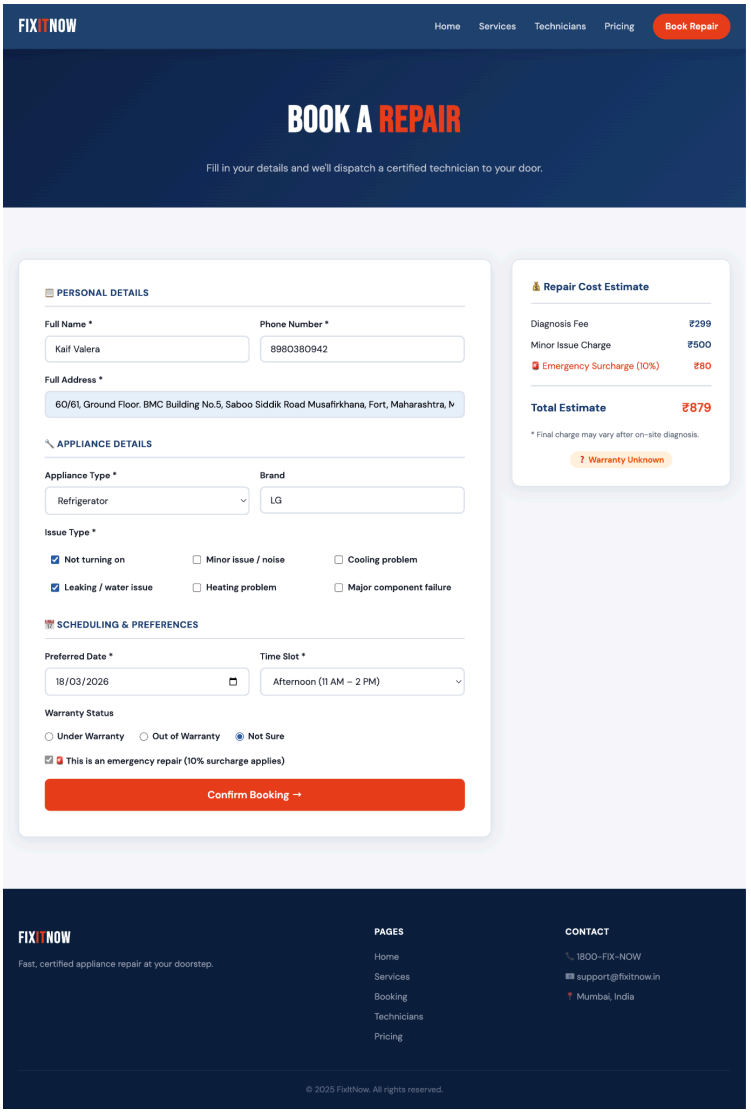
6. **style.css**

Handles all website styling and visual layout.



7. **main.js**

Handles interactive functions including form validation and cost calculation.



Key Implementation Features

- Responsive website layout
- Interactive booking form
- Dynamic repair cost estimation
- Modern UI design
- Navigation between pages

6. Problem Statement / Case Study Results and Conclusion.

Results

The FixItNow website successfully demonstrates a functional appliance repair booking platform.

The website allows users to:

- Explore repair services
- Check transparent pricing
- View technician information
- Book repair services online

JavaScript functionality ensures that:

- User inputs are validated
- Costs are calculated dynamically
- Booking confirmation is displayed after submission

Conclusion

This case study demonstrates the practical application of **HTML, CSS, and JavaScript** to develop a real-world web application.

The FixItNow system simplifies the appliance repair booking process and provides a user-friendly experience.

Future improvements could include:

- Backend database integration
- Online payment system
- User login and account management
- Real-time technician availability

Overall, the project successfully shows how front-end web technologies can be used to build a useful and interactive service booking platform.

7. References

- W3Schools HTML Documentation
- W3Schools CSS Documentation
- JavaScript Official Documentation
- MDN Web Docs
- Tutorials on Front-End Web Development